Network Protocol

School of Computer Science Engineering and Technology
Bennett University
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Outline

- ICMP
- ARP
- RARP
- BOOTP
- DHCP

Internet Control Message Protocol (ICMP)

ICMP is a network layer protocol used by network devices to diagnose network communication issues.

Used to determine the reachability of destination system in a timely manner.

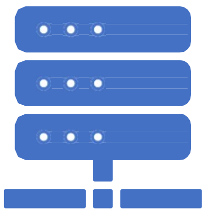
Commonly, the ICMP protocol is used on network devices, such as routers.

ICMP is used for error reporting, testing, and for distributed denial-of-service (DDoS) attacks.

ICMP messages

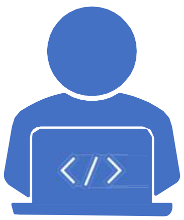
ICMP messages

Category	Type	Message
Error-reporting messages	3	Destination unreachable
	4	Source quench
	11	Time exceeded
	12	Parameter problem
	5	Redirection
Query messages	8 or 0	Echo request or reply
	13 or 14	Timestamp request or reply



Error-reporting messages:

The router encounters a problem when it processes an IP packet then it reports a message.



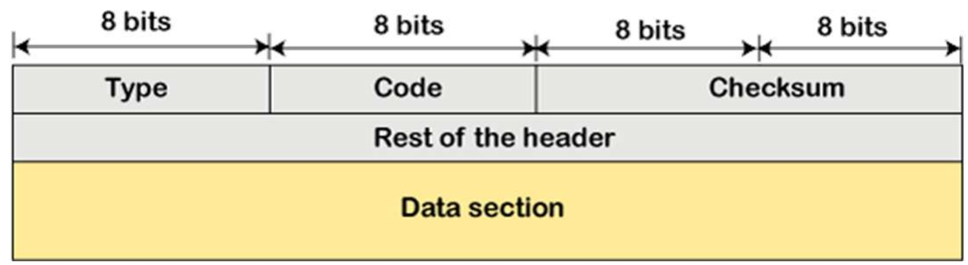
Query messages:

The messages that help the host to get the specific information of another host.

For example, suppose there are a client and a server, and the client wants to know whether the server is live or not, then it sends the ICMP message to the server.

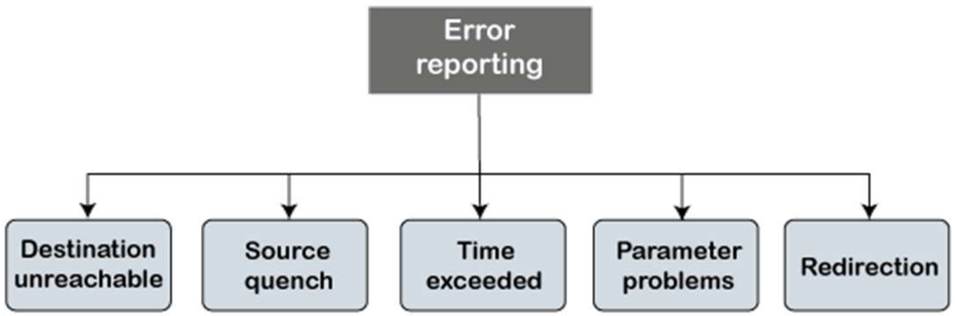
ICMP Message Format

- **The ICMP message contains the following fields:**



- **Type:** It is an 8-bit field. It defines the ICMP message type. The values range from 0 to 127 are defined for ICMPv6, and the values from 128 to 255 are the informational messages.
- **Code:** It is an 8-bit field that defines the subtype of the ICMP message
- **Checksum:** It is a 16-bit field to detect whether the error exists in the message or not.

ICMP Error Reporting Messages



- **Destination unreachable:** Inform the sender that the destination is unreachable for some reason.
- Source quench: A request to decrease the traffic rate for messages sending to the host.
- Time exceeded: Discarded datagram message sent to the sender due to time to live field reaches zero, by sending time exceeded message.
- Parameter problems: Mismatch in packet header.
- Redirection: The message informs a host to update its routing information (to send packets on an alternate route).

ICMP Query Messages

- **Echo-request and echo-reply message:**
  - A host can send an echo-request message. It is used to ping a message to another host that "Are you alive". If the other host is alive, then it sends the echo-reply message.
- **Timestamp-request and timestamp-reply message:**
  - The timestamp-request and timestamp-reply messages are also a type of query messages. Suppose the computer A wants to know the time on computer B, so it sends the timestamp-request message to computer B. The computer B responds with a timestamp-reply message.

Further studies (for students)

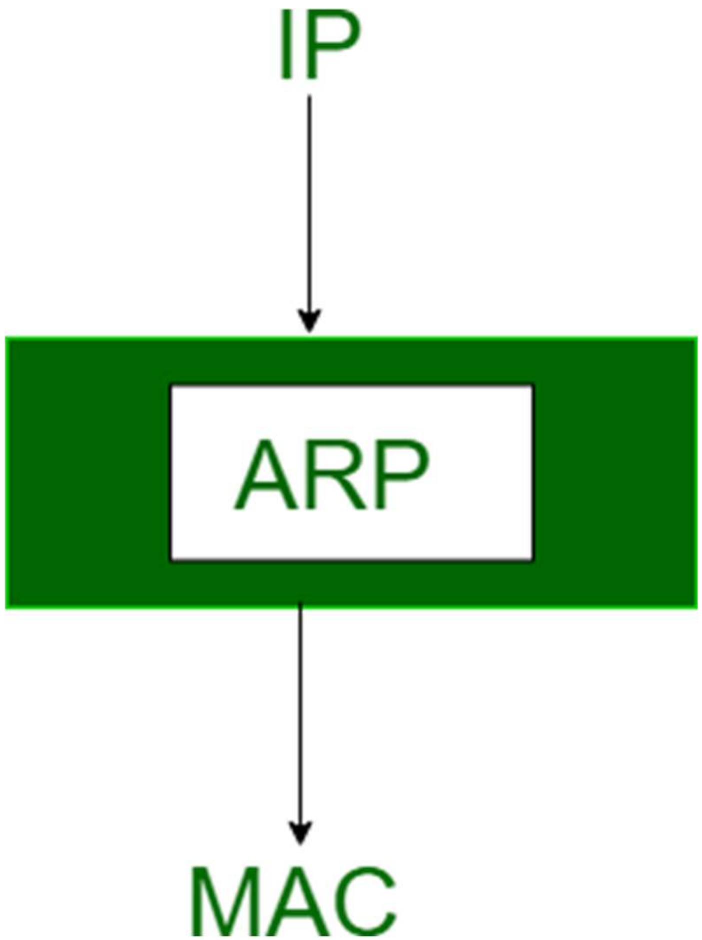
- **How is ICMP used in DDoS attacks?**
  - **ICMP Flood attack**
  - **Smurf attack**
  - **Ping of death attack**

Address Resolution Protocol (ARP)

ARP is a communication protocol used to find the MAC address of a device from its IP address.

It is used when a device wants to communicate with another device on a LAN or Ethernet.

Used to determine the reachability of destination system in a timely manner.



Address Resolution Protocol (ARP)

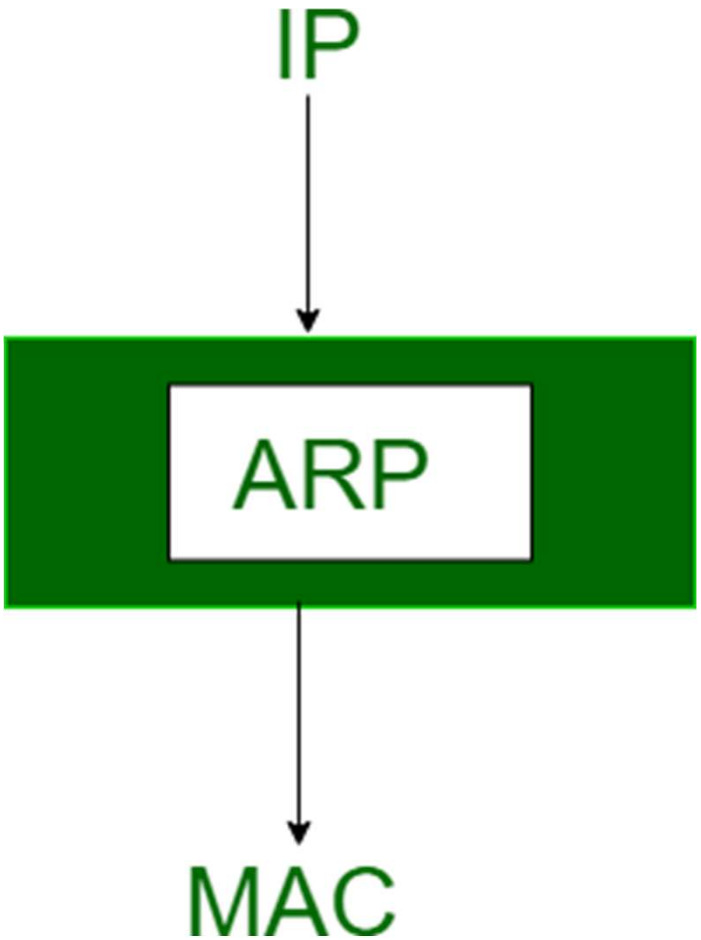
ARP is a communication protocol used to find the MAC address of a device from its Network address (IP address).

It is used when a device wants to communicate with another device on a LAN or Ethernet.

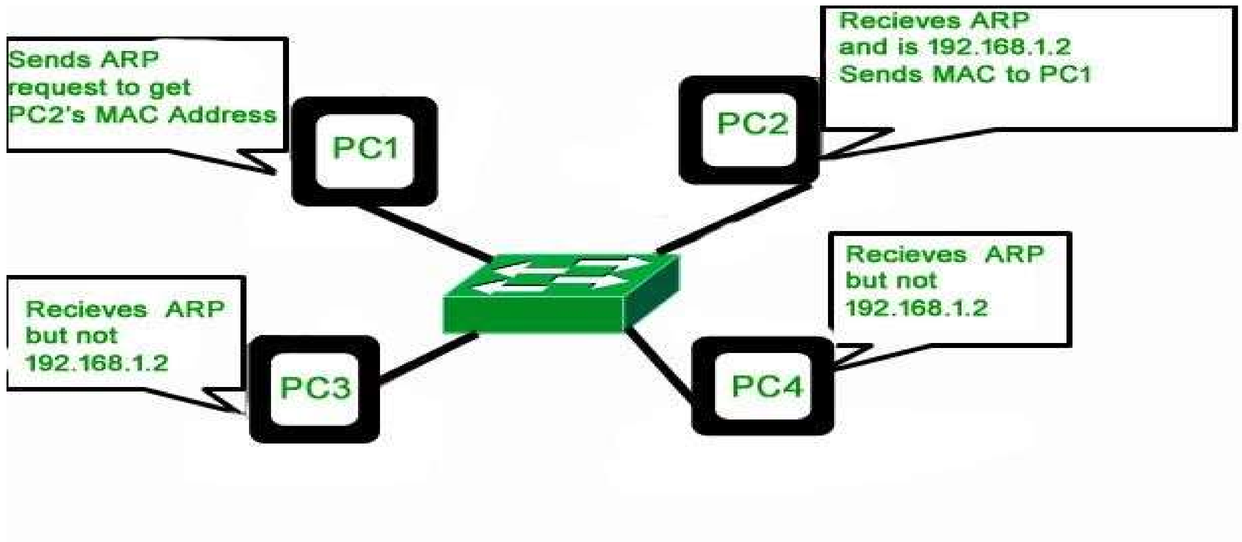
Used to determine the reachability of destination system in a timely manner.

ARP is a network layer to data link layer mapping process.

In order to send the data to destination, having IP address is necessary but not sufficient; we also need the physical address of the destination machine. .



ARP II



- Before sending the IP packet, the MAC address of destination must be known. If not so, then sender broadcasts the ARP-discovery packet requesting the MAC address of intended destination.
- ARP-discovery is broadcast, every host inside that network will get this message and discard it except that intended receiver host whose IP is associated.
- Now, this receiver will send a unicast packet with its MAC address (ARP-reply) to the sender of ARP-discovery packet. After the original sender receives the ARP-reply, it updates ARP-cache and start sending unicast message to the destination.

ARP Types

- Proxy ARP
- Gratuitous ARP
- Reverse ARP (RARP)
- Inverse ARP

Proxy ARP

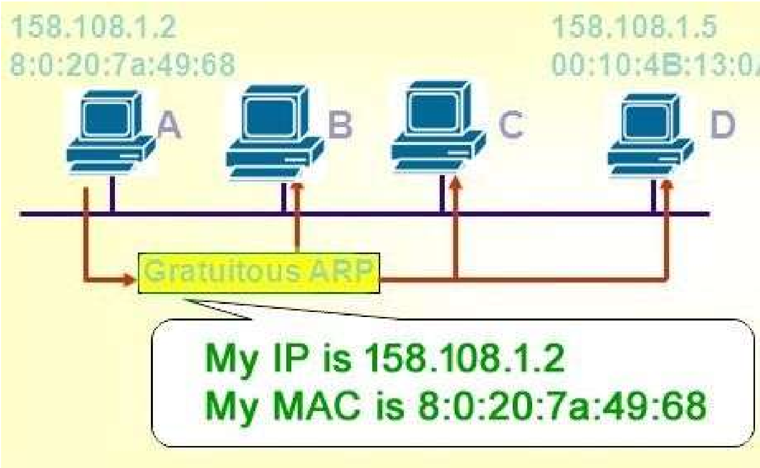
- A method through which a Layer 3 devices may respond to ARP requests for a target that is in a different network from the sender.

- The Proxy ARP configured router responds to the ARP and map the MAC address of the router with the target IP address and fool the sender that it is reached at its destination.

- At the backend, the proxy router sends its packets to the appropriate destination because the packets contain the necessary information.

  - Example - If Host A wants to transmit data to Host B, which is on the different network, then Host A sends an ARP request message to receive a MAC address for Host B. The router responds to Host A with its own MAC address pretend itself as a destination. When the data is transmitted to the destination by Host A, it will send to the gateway so that it sends to Host B. This is known as proxy ARP.

Gratuitous ARP



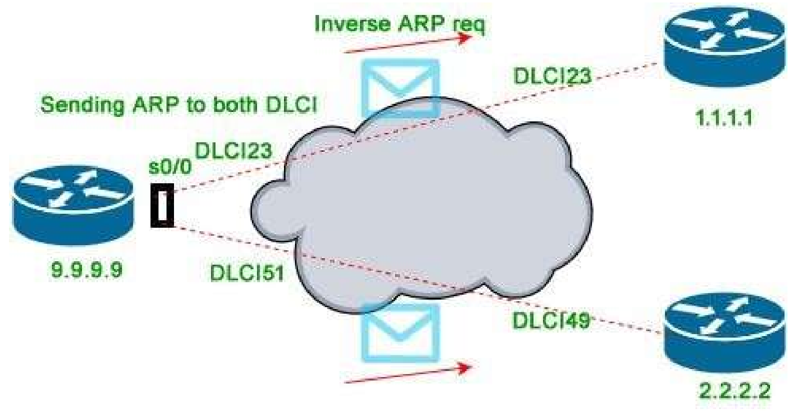
- It is used in advance network scenarios which is performed by computer while booting up. When the computer booted up (Network Interface Card is powered) for the first time, it automatically broadcast its MAC address to the entire network.
- After Gratuitous ARP MAC address of the computer is known to every switch and allow DHCP servers to know where to send the IP address if requested. Gratuitous ARP could mean both Gratuitous ARP request and Gratuitous ARP reply, but not needed is all cases.
- Gratuitous ARP request is a packet where source and destination IP are both set to IP of the machine issuing the packet and the destination MAC is the broadcast address ff:ff:ff:ff:ff:ff ; no reply packet will occur.
- It is ARP-Reply that was not prompted by an ARP-Request.
- It is useful to detect IP conflict and also used to update ARP mapping table and Switch port MAC address table.
- An ARP request of the host that helps to identify the duplicate IP address.

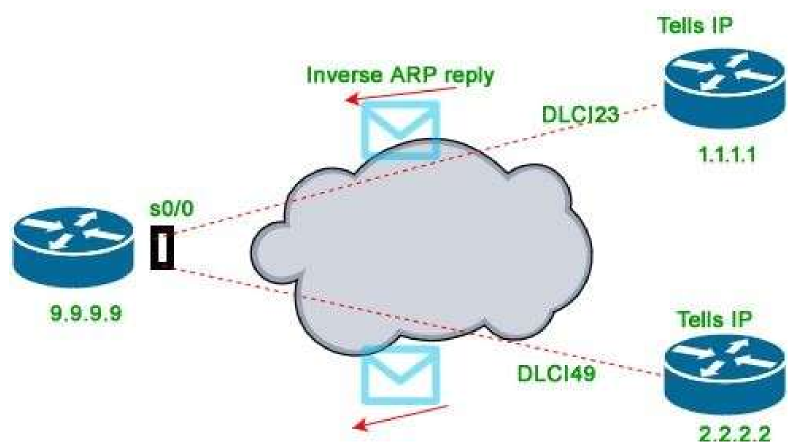
Reverse ARP (RARP)

- RARP is used by a client machine in a LAN to request its IPv4 from the gateway-router's ARP table.
- The network administrator creates a table in gateway-router, which is used to map the MAC address to corresponding IP address.
- When a new machine is setup or any machine which don't have memory to store IP address, needs an IP address for its own use. So the machine sends a RARP broadcast packet which contains its own MAC address in both sender and receiver hardware address field.
- A special host configured inside the LAN, called as RARP-server is responsible to reply for these kind of broadcast packets. Now the RARP server attempt to find out the entry in IP to MAC address mapping table.
- If any entry matches in table, RARP server send the response packet to the requesting device along with IP address.
- LAN technologies like Ethernet, Ethernet II, Token Ring and Fiber Distributed Data Interface (FDDI) support the Address Resolution Protocol.
- RARP is not being used in today's networks. Because we have much great featured protocols like BOOTP (Bootstrap Protocol) and DHCP(Dynamic Host Configuration Protocol).

Inverse ARP

- Inverse of the ARP, and it is used to find the IP addresses of the nodes from the DLL addresses.
- Used for the frame relays, and ATM networks, where Layer 2 virtual circuit addressing are often acquired from Layer 2 signaling. When using these virtual circuits, the relevant Layer 3 addresses are available.
- The InARP has a similar packet format as ARP, but operational codes are different.





Bootstrap Protocol (BOOTP)

- It is used for maintaining the protocol between connected devices on a network.
- It is a basic protocol that automatically provides each participant in a network connection with a unique IP address for identification and authentication as soon as it connects to the network. This helps the server to speed up data transfers and connection requests.
- BOOTP uses a unique IP address algorithm to provide each system on the network with a completely different IP address in a fraction of a second.
- This shortens the connection time between the server and the client. It starts the process of downloading and updating the source code even with very little information.

Bootstrap Protocol (BOOTP)

- BOOTP uses a combination of **DHCP** (Dynamic Host Configuration Protocol) and **UDP** (User Datagram Protocol) to request and receive requests from various network-connected participants and to handle their responses.
- In a BOOTP connection, the server and client just need an IP address and a gateway address to establish a successful connection. Typically, in a BOOTP network, the server and client share the same LAN, and the routers used in the network must support BOOTP bridging.
- A great example of a network with a TCP / IP configuration is the Bootstrap Protocol network. Whenever a computer on the network asks for a specific request to the server, BOOTP uses its unique IP address to quickly resolve them.

Dynamic Host Configuration Protocol (DHCP)

- A network management protocol used to dynamically assign an IP address to a device on a network so they can communicate using IP.
- DHCP automates and centrally manages these configurations, and it doesn’t require user configuration to connect to a DHCP based network.
- DHCP can be implemented on local networks as well as large enterprise networks.
- DHCP is the default protocol used by the most routers and networking equipment.
- DHCP is also called RFC (Request for comments) 2131.

Benefits of DHCP

- **Centralized administration of IP configuration:** It enables that administrator to centrally manage all IP address configuration information.
- **Dynamic host configuration:** DHCP automates the host configuration process and eliminates the need to manually configure individual host.
- **Seamless IP host configuration:** DHCP clients get accurate and timely IP configuration parameter such as IP address, subnet mask, default gateway, IP address of DND server and so on without user intervention.
- **Flexibility and scalability:** Provides increased flexibility, allowing the admin to move easily change IP configuration when the infrastructure changes.

Thank You